

IoT Challenge 3: Node-RED and LoRaWAN - Exercise

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1 EQ1

The LoRaWAN network operates at 868 MHz with bandwidth 125 kHz. The network consists of 40 sensor nodes, each transmitting 2 packets per minute. Therefore, the total traffic load is:

$$\lambda_{tot} = 40 \cdot 2 = 80 \text{ packets/min} \quad (1)$$

Converting to packets per second:

$$\lambda_{tot} = \frac{80}{60} = 1.333 \text{ packets/s} \quad (2)$$

Using the TTN airtime calculator with payload size 20 bytes, the following airtime values were obtained:

SF	Airtime (s)	$G = \lambda T$	$P_{success}$
SF7	0.0719	0.0959	82.55%
SF8	0.1336	0.1781	70.03%
SF9	0.2468	0.3291	51.78%
SF10	0.4526	0.6035	29.91%
SF11	0.9871	1.3161	7.19%
SF12	1.8104	2.4139	0.80%

The success probability is given by:

$$P_{success} = e^{-2\lambda T} \quad (3)$$

From the results, only SF7 provides a success probability greater than 75%. Therefore, the largest spreading factor that satisfies the requirement is:

$$\boxed{SF7} \quad (4)$$

2 EQ2

The most appropriate action is: Move the nodes closer to the gateway.

The measured nonuniform performance indicates that some nodes suffer from poor link quality due to excessive distance or obstacles between the nodes and the gateway. Moving the nodes closer to the gateway improves the received signal strength and increases the probability of successful packet reception.

Increasing the spreading factor would increase packet airtime and consequently increase the collision probability. Decreasing the number of nodes could reduce network load, but it would not explain why some nodes experience significantly worse performance than others.